SIYI LIU

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RESEARCH INTERESTS

1. Enhancing large-scale automation in personnel assessment with machine learning and artificial intelligence
2. Advancing understanding of how AI-delivered feedback shapes employees' experiences and behaviors at work
3. Machine learning, natural language processing, and computational modelling research methods

EDUCATION

Ph.D. **Virginia Tech, 2027**
Psychology, Major: Industrial-Organizational Psychology
Dissertation (tentative): From Micro to Macro: The (In)Efficiency of Generative AI in Organizations
Chair: Ivan Hernandez; Committee: Louis Hickman, Roseanne Foti, Chen Tang

M.S. **Virginia Tech, 2025**
Psychology, Major: Industrial-Organizational Psychology
Thesis: The Performance of Multimodal Large Language Models in Visual Personnel Assessments
Chair: Louis Hickman; Committee: Ivan Hernandez, Ning Hsu

M.S. **University of Maryland, College Park, 2023**
Finance

M.S. **Shanghai University of Finance and Economics, 2022**
Finance
Thesis: Identifying Risk Factors for Bond Default in China's Real Estate Industry Using a Random Forest Model
Advisor: Weichun Ye

B.Econ **Wuhan University of Technology, 2020**
Finance, Major: Mathematical Finance

PUBLICATIONS

Refereed Journal Articles

1. **Liu, S.**, Hickman, L., Dahlander, L., & Piezunka, H. (2026). Textual similarity in organizational research: Review of applications, consistency of methods, and best practice recommendations. *Organizational Research Methods*. https://doi.org/10.1177/10944281261432629

2. Stockdale, K.S., Hickman, L., & **Liu, S.** (2026). Scoring employment interviews with large language models: Components of structure, validity investigations, and best practice recommendations. *Journal of Applied Psychology*. https://doi.org/10.1037/apl0001396

3. Hudson, C., Hickman, L., **Liu, S.**, Young, H., Sabbagh, M., & Harkness, K. (2026). Assessing individual differences in adult theory of mind through mental state language: Human and artificial intelligence approaches. *Assessment*, 10731911261486664. https://doi.org/10.1177/10731911261486664

Book Chapters

4. Hickman, L., Stockdale, K. S., & **Liu, S.** (2026). Large Language Models in Recruitment and Selection. In I. Nikolaou, P. Dunlop, & J. McCarthy (Eds.), *Research Handbook on the Psychology of Talent Acquisition*. Preprint available at https://www.researchgate.net/publication/401691652_Hickman_Stockdale_Liu-ACCEPTED-The_Use_of_Large_Language_Models_in_Recruitment_and_Selection_chapter.

Manuscripts Under Review

5. **Liu, S.**, Hickman, L., & Liff, J. (2026). [Title withheld to preserve blind review]. [Manuscript invited for 1st revision]. *International Journal of Selection and Assessment*.

6. Hickman, L., Speer, A., Song, C., & **Liu, S.** (2026). [Title withheld to preserve blind review]. [Manuscript invited for 1st revision]. *Journal of Applied Psychology*.

7. Penedo, J. M. G., Coyne, A., **Liu, S.**, Ronchi, A., Manubens, R., Areas, M., & Hickman, L. (2026). [Title withheld to preserve blind review]. [Manuscript invited for 1st revision]. *Journal of Consulting and Clinical Psychology*.

8. Bakeman, T., Shaban T., Hickman, L., **Liu, S.**, & Rose, M. (2026). [Title withheld to preserve blind review]. [Manuscript under review]. *Journal of Applied Psychology*.

Selected Works in Progress

9. **Liu, S.**, Odusami, M.-A., & Hickman, L. (in progress). How little is enough? Examining the performance gains of fine-tuning large language models for scoring employment interview. Target: *Organizational Research Methods*. [Analysis stage]

10. **Liu, S.**, & Hickman, L. (in progress). Internalizing the algorithm? A conceptual framework of AI-delivered feedback, psychological processes, and employee Outcomes. [Editing stage]

11. Surianarayanan, V., **Liu, S.**, & Hickman, L. (in progress). Deep learning with variational rubric graph networks for uncertainty-aware selective human-AI assessment. [Analysis stage]

CONFERENCE PRESENTATIONS

Under Review

1. Liu, M. (Co-Chair) & **Liu, S.** (Co-Chair). (2027, April). *Machine learning for I-O 9.0: Pushing the LLM application frontier* [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, National Harbor, MD, United States.

2. Liu, M. (Co-Chair) & **Liu, S.** (Co-Chair). (2027, April). *Machine learning for I-O 9.0: Trust and integrity in the LLM era* [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, National Harbor, MD, United States.

3. **Liu, S.**, Odusami, M.-A., & Hickman, L. (2027, April). How little is enough? Examining the gains of fine-tuning large language models for scoring employment interview. In Wang, P. (Co-Chair) & Hou, D. X (Co-Chair), *Substitute or supplement? AI and human judgment across I-O domains* [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, National Harbor, MD, United States.

Accepted

4. **Liu, S.** (2026, September). *From Performance to Well-Being: An Integrative Conceptual Framework of AI-Powered Feedback in Organizations* [Poster presentation]. Society for Industrial and Organizational Psychology Leading Edge Consortium, Baltimore, MD, United States.

5. Hickman, L., Speer, A., Song, C., & **Liu, S.** (2026, August). *Exploring how multi-objective optimization reduces group differences in automatic essay scores* [Paper presentation]. Academy of Management Annual Conference, Philadelphia, PA, United States.

6. **Liu, S.**, Hickman, L., & Liff, J. (2026, May). The performance of multimodal large language models on visual personnel assessments: Initial evidence and implications of cheating with AI. In Campion, E. D. (Chair), *Misbehavin': Implications of AI Use in Unproctored Pre-Hire Assessments* [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, New Orleans, LA, United States.

7. **Liu, S.** (2026, February). *The performance of multimodal large language models in visual personnel assessments* [Poster presentation]. Graduate Research Symposium at William & Mary, Williamsburg, VA, United States.

8. **Liu, S.**, Hickman, L., Dahlander, L., & Piezunka, H. (2025, April). Textual similarity in organizational research: Review of applications, consistency of methods, and best practice recommendations. In Hickman, L. (Co-Chair) & Liu, M. (Co-Chair), *Machine Learning for I-O 7.0: Natural Language Processing in Research and Assessment* [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, Denver, CO, United States.

9. **Liu, S.**, Hickman, L., Dahlander, L., & Piezunka, H. (2025, April). *Textual similarity in organizational research: Review of applications, consistency of methods, and best practice recommendations.* [Paper presentation]. Annual Graduate and Professional Student Senate Research Symposium and Exposition, Virginia Tech, Blacksburg, VA, United States.

10. Stockdale, K.S., **Liu, S.**, Ansari, A., & Hickman, L. (2024, August). Scoring interviews with large language models: Initial validity evidence. In Campion, E. D. (Chair), *Applications of NLP to Human Resource Management: From Word Dictionaries to Large Language Models* [Symposium]. Academy of Management Annual Conference, Chicago, IL, United States.

11. Penedo, J. M. G., Coyne, A., **Liu, S.**, Ronchi, A., Manubens, R., Areas, M., & Hickman, L. (2024, June). *Natural language processing to identify cognitive-behavioural vs. psychodynamic interventions: Automatizing micro-analytic process evaluations in psychotherapy* [Paper presentation]. Society for Psychotherapy Research 55th International Annual Meeting, Ottawa, Canada.

12. Stockdale, K. S., **Liu, S.**, Ansari, A., & Hickman, L. (2024, April). Scoring interviews with large language models: Initial validity evidence. In Hickman, L. & Liu, M. (Chairs), *Machine learning for I-O 6.0: Applications of large language models* [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, Chicago, IL, United States.

RESEARCH EXPERIENCE

Graduate Research Assistant, Workplace Assessment and Social Perception Lab 2023 - 2026
Virginia Tech, Department of Psychology (Director: Dr. Louis Hickman) *Blacksburg, VA*

Graduate Student Researcher 2020 - 2022
Shanghai University of Finance and Economics, School of Finance *Shanghai, China*

- Developed and tuned a Random Forest classification model in thesis to predict real estate bond default risk, achieving 96.2% accuracy and 0.92 AUC; evaluated model performance using ROC analysis and implemented model interpretability techniques including Gini importance and SHAP values for feature attribution and risk factor analysis.
- Conducted an empirical quantitative finance study on Asset-Backed Notes (ABN) pricing by designing a stochastic cash-flow forecasting framework; implemented the Cox–Ingersoll–Ross (CIR) interest rate model,

Monte Carlo simulation, and Option-Adjusted Spread (OAS) methodology to simulate interest-rate paths, optimize pricing, and assess model sensitivity and risk exposure.
- Built a hybrid decision-support and performance evaluation model for a Chinese bailout fund by integrating multi-source economic, financial, and survey data; applied expert-weighted feature aggregation using the Intuitionistic Fuzzy Analytic Hierarchy Process (IFAHP) to construct interpretable composite indicators; co-authored a 40,000-word data-driven research report supporting policy and investment decision-making.

TEACHING EXPERIENCE

Guest Lecturer, Introduction to I-O Psychology Oct 2026
Virginia Tech, Department of Psychology *Blacksburg, VA*

- Delivered a guest lecture on the application of machine learning and large language models in personnel selection and assessment.

PROFESSIONAL EXPERIENCE

Graduate Consulting Intern Aug 2024 - May 2026
Virginia Tech, Department of Organizational Excellence *Blacksburg, VA*

- Collaborated with a consulting team of five to develop Virginia Tech's Organizational Excellence Model, including subscale item design; built an interactive Power BI dashboard for visualization and stakeholder use.
- Generated evidence-based policy recommendations for university leadership, derived from cross-university benchmarking and data analyses.
- Synthesized and delivered external research insights aligned with the university's strategic priorities and context.
- Coordinated with consulting team members and cross-departmental stakeholders to scope, plan, and implement university strategic projects.

Economic Affairs Intern Aug 2022 - Sept 2022
United Nations Economic and Social Commission for Asia and the Pacific, Transport Division *Remote*

- Conducted comparative literature reviews assessing sustainability progress across 12 Northeast and South Asian countries, synthesizing cross-national evidence.
- Performed qualitative policy analysis on electric mobility and sustainable transport across ESCAP member states, identifying key themes and policy patterns.
- Authored an 11,000-word analytical report on financing mechanisms and market development in Shanghai's electric mobility sector.

Overseas Investment Intern July 2022 - Aug 2022
Contemporary Amperex Technology Co. Limited (CATL), Corporate Planning Department *Ningde, China*

- Analyzed macroeconomic and industry trends affecting CATL's lithium-ion battery portfolio, synthesizing insights to support strategic decision-making.
- Maintained and updated monthly lithium market databases tracking the commercial development of 50+ global lithium mining projects, ensuring data accuracy and consistency.
- Conducted due diligence research on potential lithium acquisition targets using structured analytical frameworks (e.g., SWOT and economic analysis).
- Supported internal valuation and decision processes by compiling, validating, and organizing financial data for DCF analyses in collaboration with senior team members.

Bond Issuance Intern Jul 2021 - Oct 2021
Haitong Securities, Debt Capital Markets Department *Shanghai, China*

- Prepared and revised application and approval documents for three bond projects in Shandong Province (total value over RMB 5 billion), including bond prospectuses and advocacy reports, to meet rigorous document analysis and regulatory compliance.

- Conducted due diligence research on a target company in Renxing, China by reviewing financial and organizational information, reassessing bond eligibility, and coordinating with internal teams and external clients.
- Drafted and reviewed issuance and announcement materials, including pricing analyses; supported project tracking by updating two semi-annual corporate bond reports.
- Supported project development and team collaboration by contributing to financing proposals and pitchbooks for new bond initiatives, including preliminary issuance plans and customized solutions for issuers.

Equity Analyst Intern Feb 2021 – May 2021
CITIC Securities, Transportation Group of Research Department *Shanghai, China*

- Collected, tracked, and analyzed industry data on a daily basis to monitor key indicators, and maintained internal express, freight, and shipping databases.
- Conducted industry and policy research in the transportation sector; authored weekly and monthly research reports and analytical reviews of annual reports for listed express companies.
- Contributed to the preparation of annual report summary presentations by analyzing companies' business models, operational and financial performance; produced daily trend reports for clients.
- Participated in private interviews and strategy meetings with major industry campanies to gather up-to-date insights, synthesized findings; communicated key trends to the team leader, supporting team decision-making.

AWARDS AND HONORS

SIOP Conference Student Travel Award ($500) 2026
Society for Industrial and Organizational Psychology *New Orleans, LA*

National Graduate Scholarship in China ($3000) 2021
Shanghai University of Finance and Economics, School of Finance *Shanghai, China*

First-class Graduate Scholarship ($1700) 2021
Shanghai University of Finance and Economics, School of Finance *Shanghai, China*

Second-class Graduate Scholarship ($1100) 2020
Shanghai University of Finance and Economics, School of Finance *Shanghai, China*

Excellence Award (Selected in the Top 10%) 2020
6th National Master of Finance Case Study Competition in China *China*

First-class Undergraduate Scholarship ($420) 2019
Wuhan University of Technology, School of Economics *Wuhan, China*

Second-class Undergraduate Scholarship ($300) 2018
Wuhan University of Technology, School of Economics *Wuhan, China*

Second-class Undergraduate Scholarship ($300) 2017
Wuhan University of Technology, School of Economics *Wuhan, China*

PROFESSIONAL, INSTITUTIONAL & COMMUNITY SERVICE

Ad Hoc Reviewer
Psychotherapy Research
Organizational Research Methods
Journal of Applied Psychology, with Dr. Louis Hickman

Academy of Management
Annual Conference Reviewer, Research Methods Division (2026)

PsyCN2none: Nonprofit Psychology PhD Application Mentorship Program for Chinese Students
Mentor (2026)

Graduate Student Wellness Committee
Industrial/Organizational Area Representative, Department of Psychology, Virginia Tech (2026)

Graduate Student Study Committee
Industrial/Organizational Area Representative, Department of Psychology, Virginia Tech (2025-2026)

Tech Girls U.S. Program
Panelist at Research in STEM Fields, Virginia Tech (2025)

PROFESSIONAL REFERENCES

Dr. Louis Hickman
Assistant Professor, Department of Management, Virginia Tech
Visiting Scholar, Amazon
Senior Fellow, Wharton People Analytics, University of Pennsylvania
Email: louishickman@vt.edu

Dr. Ivan Hernandez
Associate Professor, Industrial-Organizational Psychology, Department of Psychology, Virginia Tech
Director, Computational Outreach Lab
Faculty Affiliate, Center for the Future of Work Places and Practices, Virginia Tech
Email: ivanhernandez@vt.edu

Dr. Chloe Hudson
Assistant Professor, Clinical Psychology, Department of Psychology, Virginia Tech
Director, Mental State Processing and Psychopathology Lab
Email: chloehudson@vt.edu